

**AMENDMENT NO. 1 MARCH 2005
TO
IS 4989 (PART 4) : 2003 MULTIPURPOSE AQUEOUS
FILM FORMING FOAM LIQUID CONCENTRATE FOR
EXTINGUISHING HYDROCARBON AND POLAR
SOLVENT FIRES — SPECIFICATION**

(Page 1, clause 4.1) — Delete second sentence 'It should be -----performance'.

(Page 1, clause 4.2) — Delete.

[Page 1, Table 1, Sl No. (vi)] — Substitute '*n*-heptane' for '*n*-hexane' and shift Sl No. (vi) to Table 2 on page 2.

[Page 1, Table 1, Sl No. (vii)] — Shift Sl No. (vii) to Table 2 on page 2.

(Page 1, clause 6, line 3) — Insert '50' between '30' and '100'.

(Page 2, Table 2) — Substitute the following for the existing:

(1)	(2)	(3)		(4)
With iso propyl alcohol				
i)	Expansion at $27 \pm 5^{\circ}\text{C}$	6 Min		F
ii)	25 percent drainage time at $27 \pm 5^{\circ}\text{C}$	5 Minutes		G
		<i>Polar solvent</i>	<i>n-heptane</i>	
iii)	Fire control time	120 s	60 s	H and J
iv)	Fire extinction time	300 s	90 s	H and J
v)	Burn back	10 Minutes	10 Minutes	K

(Page 2, clause 7.2) — Substitute ' 49°C ' for ' 50°C '.

(Page 2, clause 8.2) — Substitute the following for the existing:

'The product may also be marked with Standard Mark.'

(Page 2, clause A-1) — Substitute ' $27 \pm 5^{\circ}\text{C}$ ' for ' $27 \pm 1^{\circ}\text{C}$ '.

(Page 2, clause B-1.2) — Substitute ' $27 \pm 5^{\circ}\text{C}$ ' for ' 27°C '.

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[*Page 3, clause C-2.1(d)*] — Substitute the following for the existing first sentence:

'Maintain the temperature of the cooling bath at about 0°C' and in line 7, substitute '1.6°C' for '- 5°C'

(*Page 5, clause D-5.1.1*) — Substitute the following for the existing:

$$S_{a/b} = \gamma_a - \gamma_b - \gamma_l$$

(*Page 7, Annex E*) — Insert the following at the end:

E-3 CALIBRATION

E-3.1 The pressure gauges, nozzle, shall be calibrated before use when new and thereafter once every six months according to the standard procedure.

E-4 FOAM GENERATION

E-4.1 The equipment shall be arranged as shown in Fig. 3. The reservoir should be filled with 90 l of premixed solution of foam concentrate and potable water depending upon the type. The cap shall be tightened. A pressure of 8 kgf/cm² shall be built up with the help of air compressor. The discharge valve is then opened and the pressure is adjusted so as to give a pressure of 7 kgf/cm² at the inlet of the nozzle.

E-5 FILM FORMATION TEST

E-5.1 Equipment

- a) Mild steel sheet fire tray having dimension of 60 cm × 60 cm and height 15 cm.
- b) Cylindrical shaped drum made of stainless steel wire mesh having height of 45 cm, and internal diameter 20 cm. The drum is closed at lower end with M.S. sheet and opened at upper end.
- c) A suitable tong to lower and raise the drum in the tray during test.
- d) The nozzle and the arrangements as per details given in E-4.1.

E-5.2 Procedure

5.2.1 Pour water in the tray, up to height of 5 cm. Pour 2 l of *n*-heptane over the water surface. Generate the foam and operate the nozzle at 7 kgf/cm² and fill the tray by operation of the nozzle for 40 s. Immediately after stoppage of foam

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application place the cylindrical strainer drum with open end up on the tray. Wait for 2 min. Now lower a lighted torch (lighted with alcohol through the open end of the strainer and bring it near but not touching the liquid surface; care shall be exercised to ensure that burning alcohol does not drip down from the torch over the film surface. The surface shall not ignite. Now the burning torch shall be very lightly touched to the liquid surface. The surface should not ignite. Keep repeating the test at intervals of 1 min from stopping the foam application. In cases of samples having slow drainage, the film formation may be delayed and hence the sample can be considered acceptable even if it passes in the above test when tested at 5 min. The film shall hold for minimum 15 min.

NOTE — In all the above tests intermittent self-extinguishing 'walk over'/'Flash' of flame is ignored. However, if sustained burning occurs, then the film is considered weak/absent.

(Page 8, clause F-2.2) — Substitute ' $27 \pm 5^{\circ}\text{C}$ ' for ' $27 \pm 1^{\circ}\text{C}$ '

(Page 10, clause H-8) — In line 3, substitute 'jet' for 'yet' and ' $0.50 \pm 1\text{ m}$ ' for ' $0.5 \pm 1\text{ m}$ '

(Page 10, clause H-8) — In line 6, substitute '60 s' for '160 s'.

(Page 11, clause J-2) — Substitute '*n*-heptane' for 'iso propyl alcohol'

(Page 11, clause J-3) — Substitute '*n*-heptane' for 'IPA' and '3 percent' for '6 percent'